

Note on the English Translation

This English translation was prepared by the author, Rocco Martino.

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Time Series Analysis and Forecasting Methods

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Abstract

This thesis analyzes and compares several time series forecasting methods, with particular attention to classical statistical models, hybrid approaches, and alternative methods. The first part discusses simple, double, and triple exponential smoothing models (Holt-Winters), presenting their mathematical formulations, parameter estimation, and application to seasonal and nonseasonal settings. It then examines the regression framework (linear regression) and the Box-Jenkins approach, including ARIMA models and their seasonal extensions and extensions with exogenous variables (ARIMAX/SARIMAX), highlighting their effectiveness in modeling trends, seasonality, and linear dependencies.

The second part introduces more advanced approaches, such as GARCH models for forecasting heteroskedastic volatility and artificial neural networks (RNNs, LSTMs, GRUs, and hybrid architectures). It also discusses the integration of statistical and machine learning models, with particular reference to ARIMAX+ANN hybrid models and advanced HMMs, highlighting their modularity and adaptability to different types of time series.

Finally, two experiments compare selected models described in the thesis. The substantial differences between the two datasets, including the types of data they contain, provide an interesting basis for comparing the results.

The thesis focuses primarily on conceptual and applied aspects, leaving aside more rigorous mathematical implications, and provides a broad overview of the available tools and their potential for time series forecasting.

Chapter 1

Introduction and Theoretical Foundations

1.1 Introduction

This first chapter provides the conceptual tools needed to understand the discussion developed in the following chapters. Its aim is to give readers a set of fundamental concepts for interpreting the experimental results correctly and understanding the characteristics of the different stochastic approaches and techniques described here.

1.2 Problem Definition

The world around us contains many types of data. A datum can be defined as any piece of information available for forming a judgment or solving a problem.

Assigning a timestamp to each observation and arranging the values in chronological order produces a time series. The study of time series plays a fundamental role in the quantitative analysis of how phenomena evolve over time.

This thesis focuses on time series observed at regular time intervals, with a constant sampling interval. The main objective is to forecast future values of these series by studying past observations or relating them to additional variables or datasets. Forecasts are produced using different methods of analysis and interpretation and by identifying patterns, trends, and any associated seasonality. The challenge is to build models that represent the underlying data faithfully without overfitting, which limits generalization, or encountering other specific problems.

1.3 Time Series Definitions

1.3.1 Time Series

A time series is a sequence of observations ordered in time and collected at regular intervals. Formally, it is defined as a function: $\{y_t\}_{t \in T}$, where y_t is the value of the variable observed at time t , T is the set of time points, that is, the collection of observations within the period under study, and t is the time index.

1.3.2 Types of Time Series

One of the main distinctions concerns the amount of data collected at each time point. Specifically, a time series is univariate when it records a single value at time t and multivariate when it incorporates several values from different variables at the same time. Time series may also be distinguished as discrete or continuous, according to the type of data collected. For example, the price history of an individual stock is a univariate time series. A multivariate time series is obtained when temperature measurements at a location are supplemented with humidity, pressure, or other metrics.

1.4 Fundamental Principles

1.4.1 Decomposition into Individual Components

One of the main characteristics of time series is that they can be decomposed into four principal components. Each component captures a particular aspect of the value of y_t , helping us understand the series' underlying characteristics and improving our ability to analyze it. These components are:

- Trend: the long-term movement of a time series, which may be upward or downward. Essentially, it describes the overall direction followed by the data. It may also be flat or mixed and is

used to identify underlying structural dynamics within temporary movements.

- **Seasonality:** regular, recurring features of fixed duration, associated with predictable factors such as established holiday periods. In other words, its defining feature is repetition at constant time intervals.
- **Cyclical:** components similar to seasonality but without a fixed period. They may be associated with large-scale social factors, such as economic supply and demand cycles that generate cyclical market fluctuations. Cyclical components are therefore more difficult to identify because their periodicity can vary substantially over time.
- **Noise:** random, unstructured fluctuations arising from unique, nonrepeatable conditions, such as crises and extreme weather events, or from measurement errors. Noise is therefore unpredictable, and analysis seeks to filter it out or reduce its impact so that the other components become clearer.

This decomposition gives rise to two types of approach:

- **Additive** $y_t = t_t + s_t + c_t + l_t$
- **Multiplicative** $y_t = t_t \times s_t \times c_t \times l_t$

The main difference concerns how the values of the series vary over time. A multiplicative approach is preferred when growth is proportional to the level of the series; otherwise, an additive approach is preferred.

Some fundamental forecasting methods are based on moving average smoothing (MAS):

$$Y'_t = \frac{1}{m} \sum_{i=-k}^k Y_{t+i}$$

where:

- $m = 2k + 1$ is the width of the window used to calculate the average. Here, k is the distance from the central value.

This formula smooths the values of a time series using preceding and subsequent observations, centering the smoothing window on the value at time t and substantially reducing short-term fluctuations. Other forms of moving average smoothing also exist. For example, a weighted moving average (WMA) produces a smoother estimate through variable weights w , which provide additional smoothing of the variations of greatest interest. A principal approach is Seasonal-Trend Decomposition using Loess (STL), which is based on a local smoothing method called Loess. This flexible technique can model nonlinear relationships and estimate seasonal and trend components that vary over time, making it better suited to capturing complex structures than classical decomposition methods.

1.4.2 Stationarity

Before defining stationarity, it is necessary to introduce variance and autocorrelation.

Variance measures the dispersion of values around the mean and can be calculated as follows:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where:

- s^2 is the variance;
- $\bar{x}$ is the statistical mean.

Autocorrelation measures the correlation of a time series with itself shifted by a lag k , corresponding to k_t , or k previous time points.

$$\rho_k = \frac{\sum_{t=k+1}^n (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}$$

where:

- ρ_k is the autocorrelation at the k th lag;
- $\bar{x}$ is the statistical mean.
- Note that the denominator contains the total variance.

A time series is stationary when its statistical properties remain constant over time. Specifically, the mean, variance, and autocorrelation of a stationary series do not change. This is a fundamental concept when selecting a suitable forecasting algorithm. Stationarity can be further classified as weak or strict. Weak stationarity is more widely used because it is easier to verify quantitatively using the metrics described above. Stationary series are often preferred in time series analysis because they are easier to handle and forecast. Transformations such as differencing, discussed later, can be implemented in various ways to transform nonstationary time series into stationary ones. In practice, stationarity can be assessed by examining the behavior of autocorrelation functions, both the autocorrelation function (ACF) and the partial autocorrelation function (PACF). A decay as the time intervals increase may be regarded as a sign of nonstationarity. More precise statistical tests are also used, including the commonly used Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, which assesses stationarity around a deterministic trend, and the Augmented Dickey-Fuller (ADF) test, which tests the null hypothesis of a unit root, indicating nonstationarity.

1.4.3 Transformations toward Stationarity

Given the substantial influence of stationarity on the choice of forecasting algorithm, a practical and effective way to transform a nonstationary time series into a stationary one is essential for isolating the information relevant to forecasting. Several methods are available:

- Differencing: a widely used, practical principle based on calculating the difference between two consecutive observations, $Y'_t = Y_t - Y_{t-1}$. This example uses first-order differencing. Higher orders can be used when polynomial trends must be removed. A seasonal version can also be

obtained by extending the difference to the entire period s , $Y_s - Y_{s-1}$.

- **Transformations:** logarithmic transformations can be applied by taking the logarithm of the values. Their purpose is to reduce the influence of the trend and, above all, to stabilize the variance of the time series.

Differencing can be represented by a random walk model when the differenced series consists of noise, according to the formula $y_t - y_{t-1} = \varepsilon_t$, which directly yields $y_t = y_{t-1} + \varepsilon_t$. This model is used for nonstationary series in such circumstances, for example, for time series of financial prices and economic metrics.

1.5 Evaluation Metrics

1.5.1 Evaluating Forecast Accuracy

To evaluate forecasting performance, it is first necessary to distinguish between point forecasts and forecasts of probability distributions.

For point forecasts, the accuracy of the predicted term $\bar{y}(t)$ is assessed using several metrics:

- **Mean Absolute Error MAE** $= \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$
- **Root Mean Squared Error RMSE** $= \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$
- **Mean Absolute Percentage Error MAPE** $= \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|$

The first two metrics depend on the measurement scale, making comparisons between series on different scales more difficult. The last metric, MAPE, provides a percentage-based comparison and can therefore be readily used to compare time series with substantially different scales.

Other metrics are used for forecasts of probability distributions because these forecasts provide an entire distribution of values rather than a single point

value at a given time. A metric that considers only the mean is therefore insufficient: the full range and distribution of results must be analyzed. Several standard metrics are used:

- ***Quantile Score***
- ***Winkler Score***
- ***Continuous Ranked Probability Score***

The first method measures accuracy at a single quantile q . The second assesses the predictive quality of an interval constructed from multiple quantiles. Specifically, it examines the width and coverage of the prediction interval to determine whether it is too broad or too narrow.

Finally, the Continuous Ranked Probability Score (CRPS), widely used in the literature, evaluates the entire probabilistic forecast distribution. Unlike metrics based on individual quantiles or prediction intervals, CRPS considers the full predictive distribution, providing an overall measure of the model's probabilistic accuracy.

1.5.2 Residual Evaluation

One of the most important elements in assessing a forecasting method during model testing is the evaluation of residuals.

Residuals represent the remainder obtained by subtracting the predicted value from the actual value:

$$e_t = y_t - \hat{y}_t$$

Their importance lies in the possibility that they are correlated with one another. Correlated residuals indicate that the forecasting model has not adequately captured some of the available information. Similarly, if their mean is not approximately zero, the model has a bias: a remaining tendency that it has not properly incorporated.

Several indicators are used to evaluate these characteristics, including:

- ***ACF***

- ***The Box-Pierce test, defined as:***

$$Q = T \sum_{k=1}^l r_k^2$$

- The Ljung-Box test, which is more accurate for small samples, defined as:

$$Q^* = T(T+2) \sum_{k=1}^l \frac{r_k^2}{T-k}$$

The autocorrelation function (ACF) is used to reveal and clearly display correlations within the residuals. The other two indicators are primarily analytical portmanteau tests. They relate a specified lag l to the interval T , with the second being more accurate for small samples, providing an effective way to assess the nature of the residuals. Essentially, they evaluate whether the residuals may be consistent with white noise, as hypothesized. This is done by comparing the calculated statistics with threshold values from the χ_l^2 distribution with l degrees of freedom, where l is the specified lag.

Ideally, residuals should behave as white noise: they should be uncorrelated, with zero mean and constant variance.

1.5.3 Directional Accuracy

Given the considerable complexity of financial time series, traditional error metrics, such as Mean Squared Error (MSE) or Mean Absolute Error (MAE), are often insufficient or misleading. They penalize both small forecasting errors and incorrect directional forecasts, while the scale of financial series can vary substantially and inconsistently.

Directional Accuracy (DA) is the most useful indicator for evaluating predictive effectiveness on these data.

Directional Accuracy measures a model's ability to correctly predict the direction of a price movement, positive or negative, over a given period,

regardless of its magnitude. This metric is crucial for evaluating models on financial data for two main reasons:

- Statistical Edge: the directional accuracy of a random forecast corresponds to a coin toss. Any significant percentage improvement above 50% indicates that the model has captured a nonrandom statistical correlation, or edge. Even an accuracy of [percentage missing in the Italian original] represents a statistically significant advantage that can be exploited in a trading system.
- Strategic Focus: in financial markets, the fundamental trading decision is binary: buy (UP) or sell (DOWN). DA aligns directly with this decision by measuring the practical usefulness of the model for generating trading signals.

In this thesis, Directional Accuracy is the decisive metric for comparing the performance of the models in Chapter 4.

1.5.4 Additional Metrics

Other widely used metrics, although not conclusive on their own, provide useful statistics for avoiding overfitting and other problems when developing a time series forecasting or analysis system. These include:

- ***R-Squared*** (R^2):

$$R^2 = 1 - \frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{\sum_{t=1}^n (y_t - \bar{y})^2}$$

Also known as the coefficient of determination, this metric summarizes how well a model describes the input data by comparing predicted Y values, $\hat{y}_t$, with the actual test values, y_t . It is used primarily in linear regression. A

slightly different formulation, adjusted R-squared, is less sensitive to the addition of further parameters to the regression.

Another metric used in regression is:

- ***Residual Standard Error (RSE):***

$$RSE = \sqrt{\frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{n - p - 1}}$$

With p defined as the number of predictors and n as the number of observations, this metric estimates the dispersion of the average error, or residuals. In practice, it evaluates the standard deviation of the residuals.

Finally, two further fundamental metrics used in most forecasting models are AIC and its related criterion, BIC:

- ***Akaike Information Criterion (AIC)***

$$AIC = -2 \ln(\hat{L}) + 2p$$

- ***Bayesian Information Criterion (BIC)***

$$BIC = -2 \ln(\hat{L}) + p \ln(n)$$

Here, $\hat{L}$ denotes the model's maximum likelihood: the maximized likelihood of observing the specific data under the model, with p again denoting the number of parameters incorporated in the model. As the formula shows, BIC incorporates a logarithmic penalty related to the sample size n . Both criteria measure model quality and should be minimized: the model with the lowest AIC or BIC is preferred.

1.6 Visualization Methods

After calculating metrics and completing the analyses, an essential step is to visualize the resulting forecasts. Different types of charts are available, depending on the characteristic to be emphasized.

1.6.1 Time Plot

The main type of chart is the time plot, which shows how the values of the variable under consideration evolve over the time interval.

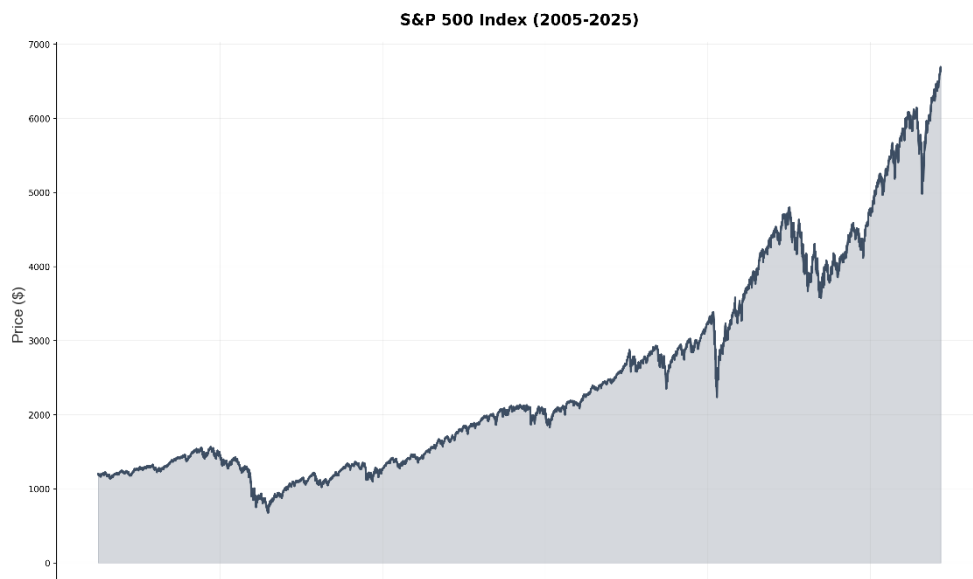


Figure 1.1: Example time plot of the S&P 500 stock index

1.6.2 Scatterplot

A scatterplot is used to visualize the relationship between two quantitative variables, highlighting correlations, patterns, or trends. For time series, it can compare the values of two series at the same time point, allowing temporal correlations between variables to be explored.

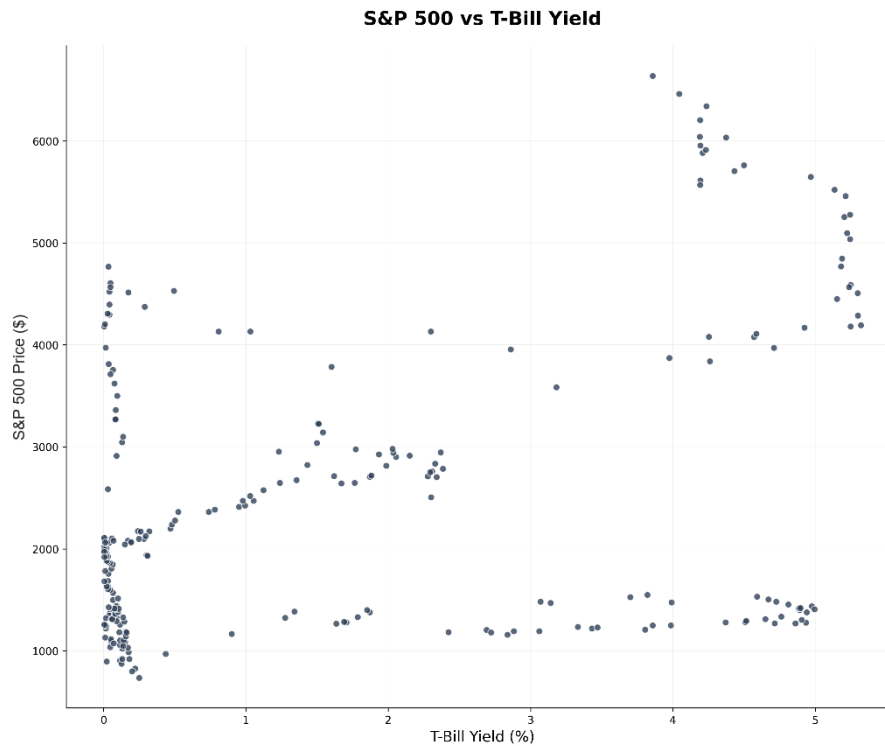


Figure 1.2: Example scatterplot showing the relationship between the S&P 500 and T-bill yield

1.6.3 Histogram

A histogram represents the distribution of a variable or a model's residuals, highlighting deviations from a reference value, often the mean.

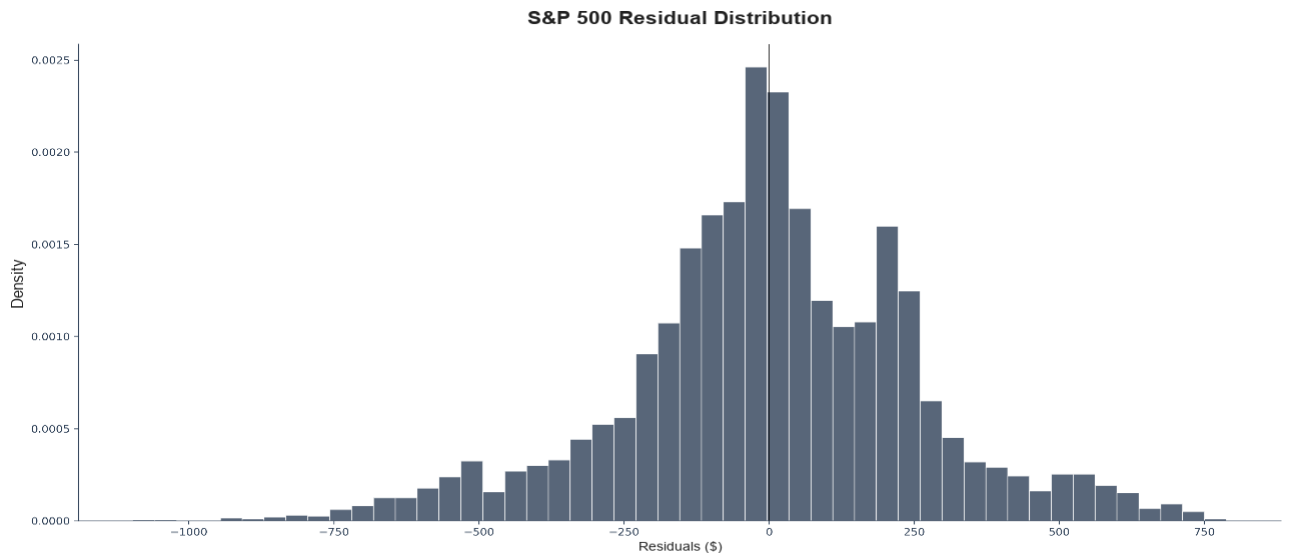


Figure 1.3: Example plot of the residuals calculated for a forecasting model

1.6.4 Correlogram

A specific visualization for the autocorrelation function (ACF) is the correlogram. It shows how variables respond at different lags and is particularly useful for identifying seasonal or cyclical patterns.

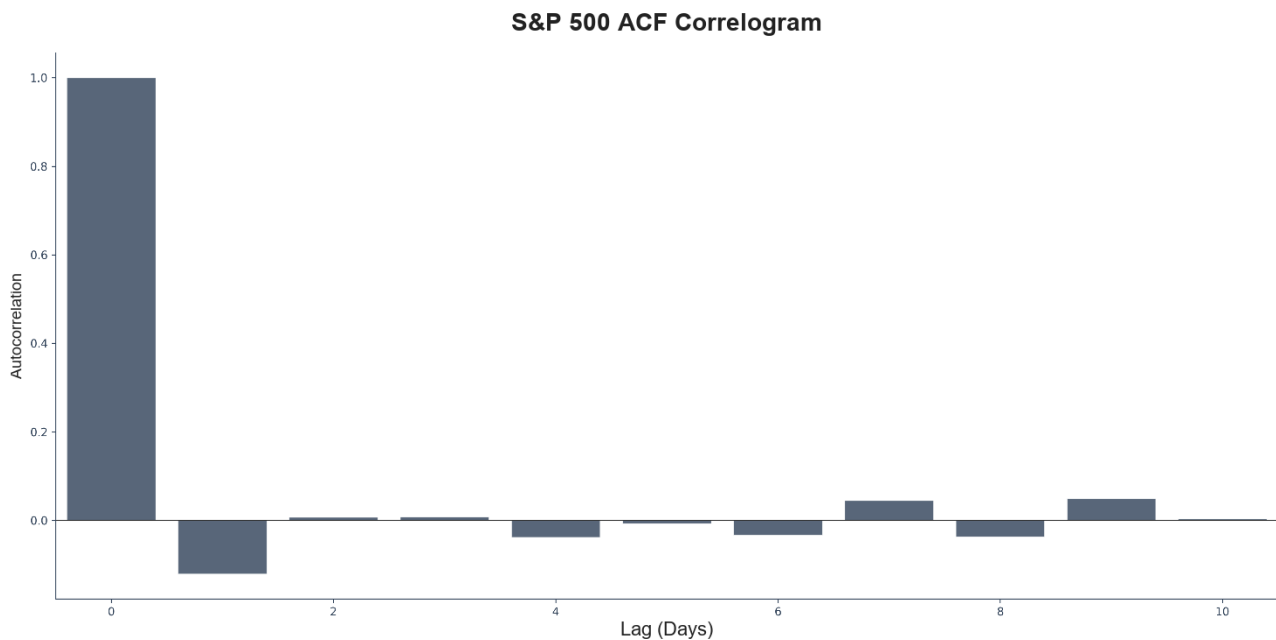


Figure 1.4: Example correlogram of the S&P 500 ACF over the period considered

Chapter 2

Classical Statistical Models

2.1 Introduction

After establishing the theoretical foundations needed to understand and analyze forecasting algorithms in the previous chapter, this section introduces several types of statistical models. It discusses methods described as classical because of their simplicity and extensive treatment in the scientific literature, rather than any lack of effectiveness.

These algorithms are widely regarded as the foundation and starting point for more advanced methods. In particular, the following are discussed:

- ***Basic forecasting methods (naive, drift, and mean)***
- ***Regression***
- ***Exponential smoothing***
- ***The Box-Jenkins approach (AR, I, MA)***

2.2 Basic Forecasting Methods

2.2.1 Mean Method

One of the most traditional approaches is the mean method. This is the simplest example considered here and is defined by the following formula:

$$\hat{y}_{T+h} = \bar{y} = \frac{y_1 + y_2 + \dots + y_T}{T}$$

The forecast $\hat{y}_{T+h}$ is the mean of the time series over the period T , at any time $T + h$, where h is the forecast horizon. This is a very simple estimate with limited informational value. Nevertheless, it is used as a starting point and benchmark when developing algorithms with greater complexity and forecasting capability.

2.2.2 Naive Method

The naive approach is another straightforward procedure, defined as:

$$\widehat{Y_{T+h}|T} = Y_T$$

The structure of this technique is also immediately apparent. The forecast $\widehat{Y_{T+h}|T}$, equivalently $\widehat{Y_{T+h}}$ relative to the period T , is obtained over the horizon h by simply setting it equal to the most recent value, the final observation in the time series. Like the previous method, this procedure is used as a baseline for subsequent developments.

Slightly different versions of the naive approach exist. For example, seasonal data can be handled by introducing the term $Y_{T+h-m(k+1)}$, where m is the length of the seasonal period and k describes the number of observations of period m within the interval $T + h$. This approach is widely used to analyze time series with seasonality.

2.2.3 Drift Method

An alternative forecasting model is the drift method, which builds on the naive approach by adding a correction coefficient:

$$\widehat{y_{T+h}} = y_t + h \left(\frac{y_T - y_1}{T - 1} \right)$$

With h again denoting the forecast horizon, an additional term is introduced relative to the standard naive method. This term is called the drift coefficient. It accounts for the change between the first and last observations within the interval considered, divided by the number of observations. The method's main objective is to estimate and incorporate a constant average rate of change, or drift, that adds fixed increases or decreases to the forecast values. Essentially, it calculates the average rate of change per period and systematically projects it into the future.

This technique is particularly effective and widely used for time series that satisfy two main conditions:

1. They have no substantial cyclical or seasonal components.
2. They display clear, monotonic trends, either increasing or decreasing.

The drift method transforms a trend that might otherwise be difficult to forecast consistently into a straight line with a constant slope, making it easier to handle and extrapolate over the forecast horizon.

2.3 Time Series Regression

2.3.1 Introduction

Regression is based on the central idea that the y values of the time series of interest are related to auxiliary variables, also called predictors. In linear regression, this is a first-order relationship. Analyzing and modeling that relationship makes it possible to estimate effective parameters that form a predictive structure for the dependent variable y .

2.3.2 Simple Linear Regression

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

In simple linear regression, the coefficients to be estimated are β_0 and β_1 , together with the error ε_t . The term x_t is the auxiliary variable. The objective is to estimate these coefficients quickly and effectively; their calculation is discussed below.

Similarly, logistic regression estimates coefficients with a comparable structure, but applies them through a logistic link function to model the probability of membership in a particular category, such as the positive or negative direction of a series. Linear regression is used to estimate continuous variables, whereas logistic regression is better suited to binary classification or the prediction of discrete events, providing a natural extension of the regression framework to a probabilistic setting.

2.3.3 Multiple Linear Regression

$$y_t = \alpha + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \dots + \beta_k x_{k,t} + \varepsilon_t$$

This second approach is a straightforward generalization of the simple case, relating y_t to k auxiliary variables, $x_{k,t}$. The original objective remains unchanged: to estimate the beta coefficients and the error ε_t .

When selecting predictors, it is important to consider potential problems. First, a definite relationship between the predictors and the variable to be estimated is assumed; specifically, each predictor x is assumed not to be a random variable. Second, the relationship is assumed to be linear. This latter assumption is particularly important because the coefficient estimation methods are based on it.

2.3.4 Least Squares Method

The standard approach for estimating the beta coefficients is Ordinary Least Squares (OLS), defined by minimizing the following expression:

$$\sum_{t=1}^T \varepsilon_t^2 = \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t} - \dots - \beta_k x_{k,t})^2$$

Given the values of the variables $y_t, x_{1,t}, \dots, x_{k,t}$, fitting searches for the optimal values that minimize the expression, specifically the error ε_t . The aim is to find the line, plane, or hyperplane in the case of multiple variables that comes as close as possible to all observed points, minimizing the sum of squared differences between actual values and model predictions. This can be achieved using widely available algorithms integrated into software applications.

2.3.5 Estimating the Dependent Variable

The dependent variable y is estimated using the following calculation:

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_{\{1,t\}} + \hat{\beta}_2 x_{\{2,t\}} + \dots + \hat{\beta}_k x_{\{k,t\}}$$

The coefficients $\hat{\beta}_{1,\dots,k}$ are the optimal values obtained by applying OLS. Substituting the values of the auxiliary variables x into the equation produces an estimate of the dependent variable y , denoted by $\hat{y}_t$. The quality of the estimate can be assessed quantitatively using R^2 , as discussed earlier, or more precisely using adjusted R^2 . The latter accounts for the number of variables

in the model, enabling a more accurate assessment of the actual improvement in explaining the variability of y_t and avoiding an overestimation of the contribution of additional variables with little informational value.

Another previously discussed metric involves calculating residuals from the following formulation of the error:

$$\begin{aligned} e_t &= y_t - \hat{y}_t \\ &= y_t - \hat{\beta}_0 + \hat{\beta}_1 x_{1,t} + \hat{\beta}_2 x_{2,t} + \dots + \hat{\beta}_k x_{k,t} \end{aligned}$$

Using the ACF and residual plots, we can assess whether the residuals still contain information that the linear regression model has not fully captured. Together with additional tests such as Ljung-Box, these metrics enable a detailed diagnosis of model results.

2.3.6 Selecting Auxiliary Variables

An equally central issue is the quantitative selection of auxiliary variables to include in the model, particularly when there are many independent variables or strong correlations between them.

Classical methods such as Best Subset Selection (BSS) and Forward Stepwise Selection (FSS) have long been considered standard strategies. However, recent simulations have highlighted limitations in their performance. Although BSS always includes the correct model among its candidate solutions, it tends to perform poorly in realistic scenarios with moderate or high correlations between predictors and low signal-to-noise ratios. Its high computational cost also makes BSS impractical at scale. FSS is simpler, but produces results very similar to BSS without offering substantial advantages.

Penalized methods such as Lasso and Elastic Net (Enet) have shown better performance, particularly in high-dimensional settings with correlated predictors and weak signals. They produce more parsimonious and stable models, reducing the risk of including auxiliary variables that do not provide useful information. Of the two, Enet is particularly suitable when predictors are strongly correlated because it combines penalties based on the norms of the beta coefficients: L_1 , which sums their absolute values and forces some

coefficients to zero, automatically selecting variables, and L_2 , which sums their squares and reduces their weight rather than selecting them.

In summary, the L_0 norm penalty, which considers the number of nonzero beta coefficients and is typical of BSS, is suitable only under ideal conditions with independent predictors and very strong signals, circumstances that are uncommon in real data. In more complex applications, strategies such as Elastic Net with Ridge-type weights provide a more robust and effective choice, balancing interpretability, accuracy, and model stability.

<i>Norm</i>	<i>Formulation</i>	<i>Associated Method</i>	<i>Considerations</i>
L_0	$\ \beta\ _0 = \sum_{j=1}^p I(\beta_j \neq 0)$	<i>BSS</i>	Counts nonzero variables, requiring discrete selection (an NP-hard problem)
L_1	$\ \beta\ _1 = \sum_{j=1}^p \beta_j $	<i>Lasso</i>	Promotes sparse solutions: some coefficients are exactly zero
L_2	$\ \beta\ _2^2 = \sum_{j=1}^p \beta_j^2$	<i>Ridge</i>	Penalizes large coefficients and stabilizes estimation without setting them to zero

The L_1 penalty is not differentiable at zero. Geometrically, the L_1 constraint is diamond-shaped, with corners on the axes, and the optimization solution tends to fall on those corners. As a result, some coefficients become exactly zero. This is why Lasso automatically selects variables.

2.4 Exponential Smoothing

2.4.1 Introduction

Exponential smoothing refers to a set of techniques and methods that calculate a weighted average of past values, applying exponential decay to earlier observations by progressively reducing their weights.

2.4.2 Simple Exponential Smoothing

The most conventional and easily implemented variant is simple exponential smoothing (SES), first discussed in the statistical literature by Brown. It is of type (N, N), meaning that the time series to be forecast has neither trend nor seasonality:

$$\widehat{y_{T+1}} = \sum_{j=0}^{T-1} \alpha(1-\alpha)^j y_{T-j} + (1-\alpha)^T \ell_0$$

Here, $\alpha \in [0,1]$ is the smoothing coefficient, while ℓ_0 is the initial level of the series. For large periods T , the entire term involving the initial level is often omitted. Both coefficients must be estimated. Starting with the first iteration, the estimate of $\hat{y}_0$, the formula uses a recursive approach to obtain a forecast of the future value $\widehat{y_{T+1}}$. The main difficulty lies in calculating the coefficients. Unlike conventional regression, the equation is nonlinear, so analytical equations for finding the optimal solutions are unavailable.

The same sum of squared errors minimization problem applies:

$$SSE = \sum_{t=1}^T (y_t - \hat{y}_t)^2 = \sum_{t=1}^T e^t$$

Numerical optimizers are therefore required to estimate α and ℓ_0 . These methods are described at the end of the theoretical discussion.

This initial formulation is widely used for the many time series without trends or seasonality, such as financial series and similar data.

2.4.3 Double Exponential Smoothing

The double approach essentially adds a further parameter, beta, which models trends within the time series. First described by Holt, this method is formulated for forecasts over a horizon h as follows:

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$\widehat{y_{t+h}} = l_t + h \cdot b_t$$

This second formulation introduces an additional coefficient, beta, used to model time series with linear trends. The coefficient β is also defined over the interval $[0,1]$. Low values of β indicate a slower trend with fewer fluctuations over time, whereas high values estimate faster trends that are more sensitive to changes in the growth rate.

Initial values must also be chosen in this case, as in the previous case with ℓ_0 . The additional parameter b_0 is generally set equal to the difference in the variable between t_2 and t_1 .

An alternative way to define the trend is the damped trend approach, developed after Holt's original method. It introduces an additional variable, phi, also $\in [0,1]$, called the trend damping coefficient. This coefficient flattens Holt's trend, particularly over the long term, to prevent unrealistic overestimation.

2.4.4 Triple Exponential Smoothing

A further formulation is triple exponential smoothing, known as the Holt-Winters method. It introduces a third variable into the model to capture the seasonality of the time series being analyzed:

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$s_t = \gamma(y_t - l_t) + (1 - \gamma)s_{t-m}$$

Future forecasts can be expressed using two alternative approaches:

- **Additive** $\widehat{y_{t+h}} = l_t + h \cdot b_t + s_{t-m+h_m}$
- **Multiplicative** $\widehat{y_{t+h}} = (l_t + h \cdot b_t) \cdot s_{t-m+h_m}$

Three new variables are introduced. The new coefficient γ , like α and β , always lies within $[0,1]$ and models the seasonality of the series. The term m is the length of the seasonal cycle, and s_t is the seasonal component included in the forecasting equation. The term h_m is an index used to repeat the seasonal pattern over the forecast horizon h .

Two approaches are generally adopted: additive and multiplicative. The multiplicative approach is appropriate when the amplitude of the seasonal component varies in proportion to the level of the time series, so that seasonal intensity increases or decreases as the level changes. The additive approach is used when seasonal amplitude remains constant regardless of the level of the series. It is therefore more suitable for phenomena whose seasonal fluctuations have a uniform magnitude over time.

2.4.5 ETS Modeling

The preceding discussion can be brought together into a family of models commonly referred to as Error-Trend-Seasonal (ETS), where:

- E (Error): the error component may be additive (A) or multiplicative (M).
- T (Trend): the trend component may be absent (N), additive (A), or damped (A_d).
- S (Seasonality): the seasonal component may be absent (N), additive (A), or multiplicative (M).

The general formulation can be expressed as:

$$y_t = f(l_t, b_t, s_t) + \varepsilon_t$$

This representation makes it possible to construct models tailored to the specific characteristics of the time series being analyzed. The strength of ETS lies in this flexibility. Its intuitive and effective structure allows the available combinations to be selected according to forecasting performance. Selection typically involves comparing models using information criteria such as AIC, BIC, and AIC_c , introduced in Chapter 1.

These properties make ETS models one of the most widely used methods in applied time series analysis and forecasting.

2.4.6 Parameter Estimation

As noted earlier, estimation of the parameters (α, β, γ and ϕ) and initial values in the ETS framework is based on minimizing the Sum of Squared Errors (SSE). The nonlinear nature of the recursive smoothing equations prevents an analytical solution to this optimization problem. Iterative numerical optimization algorithms are therefore needed to find minimum values within the constrained parameter space.

These algorithms can be divided into two main categories:

- **Derivative-Free Methods (Nelder-Mead):** the Nelder-Mead algorithm is a direct, derivative-free method that iteratively deforms a simplex in the parameter space. Its main strength is robustness, making it particularly suitable for finding minima of complex functions or identifying a valid starting point.
- **Quasi-Newton Methods (BFGS):** quasi-Newton methods, such as BFGS and newer variants such as HBFSGS, are preferred for their convergence efficiency. These algorithms approximate the Hessian matrix using gradient information, often calculated numerically, providing substantially faster convergence.

Optimization is often performed using a multistart strategy, in which minimization is repeated from several initial points. This approach mitigates the risk of convergence to a local optimum, helping the final parameter estimates approach the global minimum of the SSE as closely as possible.

2.5 The Box-Jenkins Approach (ARIMA)

2.5.1 Introduction

The Box-Jenkins approach is an operational framework for formalizing and applying ARIMA forecasting models. It provides a systematic process for constructing, estimating, and evaluating these models so that they can adequately describe and forecast a time series.

2.5.2 AR, I, and MA

Autoregressive Integrated Moving Average (ARIMA) identifies a class of models that can generally be expressed by the following equation, comprising three main components:

$$\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t$$

where:

- $\phi(B) = (1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p)$ is the autoregressive (AR) polynomial of order p ;
- $(1 - B)^d$ is the differencing (I) operator of order d ;
- $\theta(B) = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q)$ is the moving average (MA) polynomial of order q .

In this formulation, B is the backshift operator applied to the time series index, such that $B^j X_t = X_{t-j}$: it shifts the series back by j periods. The most general ARIMA formulation also includes a constant term c . For a series that is already stationary ($d = 0$), this represents the starting point of the ARMA model. When $d = 1$, it acts as drift, a constant linear trend. For values of $d > 1$, the interpretation of the constant is less immediate, and the term is often omitted, implicitly assuming that the differenced series has zero mean.

The model components describe the following:

- AR Component: the autoregressive polynomial describes the dependence of a time series on its past values. The order p specifies the number of lags considered. In other words, the current value of the series is expressed as a linear combination of previous values.
- I Component: the differencing operator can transform a nonstationary series into a stationary one. The order d is the number of times the series must be differenced to remove a trend or deterministic seasonality.
- MA Component: the moving average polynomial describes the dependence of the current variable on errors made in forecasts for preceding periods. The order q specifies the number of error terms included in the model, capturing how past errors contribute to the current value of the series.

ARIMA can therefore be considered a generalization of ARMA. When a time series is stationary from the outset, differencing is unnecessary, and the model

reduces to ARMA, focusing exclusively on the autoregressive and moving average components.

2.5.3 Operational Methodology

To make this modeling approach effective in practice, an operational procedure must be defined. Box-Jenkins formalizes precisely this procedure.

It can be summarized in four steps, with the first three repeated iteratively:

I. **Identification:**

The first step is to assess the stationarity of the series and determine the differencing order d using the statistical tests discussed earlier, including ADF and KPSS. The orders p and q are then estimated by analyzing the autocorrelation function (ACF) and partial autocorrelation function (PACF). Sets of candidate (p, d, q) triples are selected at this stage for subsequent evaluation.

II. **Parameter Estimation:**

The parameters ϕ_p and θ_q are estimated using numerical optimization methods, typically based on maximum likelihood or iterative algorithms that minimize the sum of squared errors.

III. **Model Diagnostics:**

Residuals are analyzed to assess the adequacy of the estimated model. Tools such as the residual ACF and the Ljung-Box and Box-Pierce tests are used to determine whether the residuals behave as white noise, with no significant autocorrelation and constant variance. If they do not, the process returns to identification to redefine the model orders.

IV. Forecasting:

After iterating through steps 1-3, a set of valid models is obtained. Selection among them is based on AIC, BIC, and AIC_c , where a lower value indicates a more parsimonious model with better forecasting capability. Once the optimal variant has been selected, it can be used to forecast the time series.

2.5.4 ARIMA Extensions

Although the ARIMA family is robust, extensions are needed to handle more complex data structures, particularly seasonality.

One of the most significant variants is SARIMA (Seasonal Autoregressive Integrated Moving Average). As its name suggests, SARIMA is specifically designed to forecast time series with a clear seasonal component, as described earlier.

SARIMA captures seasonal information by introducing three additional parameters, P, D, and Q, alongside the nonseasonal parameters:

- $(P, D, Q)_s$: P, D, and Q are the seasonal orders of the autoregressive, integrated (differencing), and moving average components, and s is the seasonal period.

The inclusion of these seasonal parameters reflects the modular nature of the standard ARIMA model. The ability to adapt and add components to explain specific phenomena, such as seasonality, without changing the fundamental AR(p), I(d), and MA(q) structure is a distinctive feature. It enables ARIMA to be integrated effectively into various advanced forecasting systems, some of which are discussed in Chapter 3.

Chapter 3

Hybrid Models and New Approaches

3.1 Introduction

This final theoretical chapter presents hybrid models and more recently developed approaches to time series analysis and forecasting. Its objective is to provide an overview of methods that integrate classical models with innovative techniques to overcome their limitations and improve forecasting capability.

The discussion does not examine mathematical details, which are often considerably complex. Instead, it provides a conceptual and practical understanding of these tools, highlighting their operating principles and application contexts. In particular, it examines the principal hybrid strategies and the contributions of advanced statistical tools and machine learning methods.

3.2 Dynamic Regression Models (ARIMAX)

Although the ARIMA family and its seasonal variant, SARIMA, provide a robust framework for univariate modeling, they assume that a time series' future depends exclusively on its own past. Many economic and other real-world phenomena, however, are significantly influenced by external factors or other time series.

To overcome this limitation, the ARIMAX model (Autoregressive Integrated Moving Average with Exogenous Regressors), also known as Reg-ARIMA, is used.

3.2.1 Structure

As the model's name suggests, auxiliary, or exogenous, variables are integrated into the standard ARIMA approach, dividing its operational methodology into two components:

- **Deterministic Component:** essentially a linear regression that captures the predictive contribution to y_t from the exogenous variables $(x_{1,t}, \dots, x_{k,t})$.
- **Stochastic Component:** the ARIMA model explains the residuals and the structure of the noise, encompassing aspects that cannot be attributed to external factors.

This two-component structure improves estimation and adaptability to complex datasets that depend on additional variables.

3.2.2 Formulation

The ARIMAX(p, d, q) approach is defined as follows:

$$\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t + \boldsymbol{\beta}(B)\mathbf{X}_t$$

where:

- $\phi(B)(1 - B)^d y_t = \theta(B)\varepsilon_t$ corresponds to the standard ARIMA model;
- $\boldsymbol{\beta}(B)\mathbf{X}_t$ is the regression term that captures the linear relationship between the exogenous variables and the series y_t .

For k exogenous variables, the term is replaced by the following sum: $\sum_{i=1}^k \beta_i x_{i,t}$. ARIMAX combines the characteristics of multiple regression with the autoregressive and moving average components of standard ARIMA models. This combination highlights the model's modularity: it can capture both relationships with exogenous factors and the internal dynamics of the time series. This integration is ARIMAX's main strength. Each component emphasizes a specific aspect of the data, and combining them produces forecasts that are more robust and better reflect the complexity of the phenomenon being analyzed.

3.2.3 Assessment of ARIMAX

This hybrid approach is widely used in econometric and quantitative analysis, where much of the variation observed in time series also depends on external explanatory variables. Its applicability is not limited to economics: it can be extended to any phenomenon that is correlated with external variables. The model can also be readily generalized to a seasonal formulation, yielding SARIMAX, which jointly handles trends, seasonality, and exogenous factors. These characteristics make ARIMAX and its seasonal extension, SARIMAX, among the most extensively studied and applied modern approaches to time series analysis.

3.3 The GARCH Approach

ARMA and ARIMA models are designed to model the conditional mean of a time series: the expected value of the variable at time t given past observations and errors. They implicitly assume that the variance of forecast errors (ε_t) remains constant over time, a property known as homoskedasticity. In several fields, however, including financial and macroeconomic data analysis, this assumption is almost always violated.

3.3.1 Heteroskedasticity

Heteroskedasticity is the property of a data series whose forecast error variance is not constant but changes over time, often forming volatility clusters.

During periods of high volatility, more substantial positive or negative changes are more likely, so large variations in values are easier to observe. Conversely, during periods of low volatility, substantial changes are much less frequent. Conditional heteroskedasticity applies when the variance depends systematically on past errors or variances: it follows a dynamic model and therefore has a structure.

This is widely applicable to quantitative finance, where the volatility of a financial instrument can be clearly identified.

GARCH is based on this characteristic and aims to estimate and evaluate time series that exhibit it.

3.3.2 GARCH Model Formulation

GARCH (Generalized Autoregressive Conditional Heteroskedasticity) generalizes the earlier ARCH model. The distinction described here is that GARCH explicitly models the conditional variance of the series.

GARCH treats the conditional variance (σ_t^2) as though it were itself an ARMA process. A typical GARCH(p, q) model is defined by the following equations:

- **Mean Equation**

The term r_t denotes the expected return, while the error ε_t is represented by the equation into which the calculated variance is substituted.

$$r_t = \mu_t + \varepsilon_t$$

$$\varepsilon_t = \sigma_t z_t \text{ with } z_t \sim N(0,1)$$

where:

μ_t is the conditional mean, often estimated using an ARMA model, and σ_t is the conditional standard deviation, the square root of the variance defined below.

- **Variance Equation**

This is the central equation of the GARCH(p, q) model, describing the evolution of the conditional variance (σ_t^2):

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Two fundamental components can be identified, aside from ω , the intercept or initial value. The first, the ARCH term, models dependence on past squared errors, ε_{t-i}^2 , capturing the immediate effect of the error on the time-dependent value of the series. The second, the GARCH term, incorporates past conditional variance into the model, adjusting values for remaining historical inertia and thus modeling volatility clustering.

The values of α_i and β_j , as in other approaches discussed here, must be estimated by maximizing the likelihood function, described as MSE in the original text, or using numerical optimization algorithms that minimize the log-likelihood function.

3.3.3 Concluding Remarks on GARCH

GARCH is a standard tool for transforming volatility, traditionally a source of inefficiency and uncertainty in forecasting, into a process that can be modeled and predicted. This is crucial for constructing more reliable estimates, improving quantitative financial strategies, and supporting decisions based on risk assessment.

GARCH's flexibility has also led to numerous extensions, including EGARCH, TGARCH, and GJR-GARCH, developed to capture specific phenomena such as asymmetric responses to shocks and leverage effects. Whereas ETS and ARIMA models focus primarily on the conditional mean of a time series and its linear structure, GARCH addresses an equally crucial aspect: conditional variance.

Combining different models, such as ARIMA and GARCH, illustrates the modularity of time series analysis and is widely practiced in the applied literature. GARCH models are a fundamental part of this set of methods, marking a transition from a purely linear approach toward a perspective that captures the inherent complexity and nonlinear components of real-world data.

3.4 Neural Networks for Time Series

3.4.1 Introduction

Although classical linear models, such as ARIMA and GARCH, provide a robust framework for time series analysis, their ability to identify and model complex nonlinear relationships, often present in economic and financial data, is limited. Artificial Neural Networks (ANNs) and deep learning offer a powerful nonparametric alternative because they learn directly from data without imposing a rigid functional structure. A detailed mathematical treatment of these models is beyond the scope of this work. This section provides only an introductory overview of their general principles and potential applications.

3.4.2 Fundamental Neural Network Architecture

These neural models were originally inspired by the operation of the human brain. Artificial neurons process input signals and produce outputs that are passed to subsequent layers of neurons. This multilayer architecture represents relationships between variables nonlinearly, progressively adapting to the complexity of the available data. Specifically, it maps a relationship between X_t (input) and $\hat{Y}_t$ (output) without defining a fixed formula or relationship in advance. The objective remains to forecast the future value of y_t from past values (y_{t-1} , y_{t-2} , ...) and, in some cases, exogenous variables (x_t).

A standard approach is the feedforward neural network (FFNN), specifically the Multilayer Perceptron (MLP), which consists of:

- Input Layer: time series values, including lagged values, or exogenous variables.
- Hidden Layers: the layers in which relationships are formed through neurons and activation functions, such as the sigmoid function.

- Output Layer: the layer that produces the forecast $\hat{Y}_t$.

Thus, $\hat{Y}_t$ is defined by:

$$\hat{Y}_t = f\left(\sum_i x_i w_i + b\right)$$

Here, w_i are the weights, representing the importance assigned to each input, analogous to coefficients in an ARMA model, and b denotes the biases. Learning consists of iteratively adjusting weights and biases to minimize a cost function, typically Mean Squared Error (MSE) or its square root (RMSE), as discussed in the next section. The term $f(\cdot)$ is the activation function, which introduces nonlinearity. This creates an interesting parallel: the weights w_i play a role similar to coefficients in ARIMA or ARIMAX, but in ANNs they are not restricted to linear relationships. Instead, they are combined with nonlinear functions that can model more complex dynamics.

3.4.3 Learning and Optimization Algorithms

The operation of an artificial neural network centers on learning, or training: the model's ability to adapt its internal parameters, the weights w_i and biases b , based on observed data. The objective is to reduce the discrepancy between predicted values $\hat{Y}_t$ and actual values y_t by minimizing a cost function that measures forecast error.

The most commonly used cost functions in time series applications are Mean Squared Error (MSE) and its square root (RMSE), discussed and defined in Chapter 1.

Cost minimization uses an iterative mechanism known as backpropagation, which efficiently calculates the gradients of the cost function with respect to the network parameters. These gradients are then used to update weights and biases according to a gradient descent rule.

The main gradient descent algorithms differ primarily in their parameter update strategies. Stochastic Gradient Descent (SGD) updates weights using small subsets of data, helping avoid entrapment in local minima. Adam combines momentum, or trend following, with adaptive learning rates to accelerate convergence. RMSProp automatically adjusts the update step according to gradient magnitude, improving training stability.

These algorithms allow the neural network to progressively model relationships between inputs and outputs, including complex nonlinear relationships.

A fundamental requirement is to avoid overfitting: excessive adherence to historical data that compromises forecasting performance on future data. Regularization techniques, such as dropout and weight decay, and cross-validation methods are therefore used to evaluate performance on data excluded from training.

The learning process in neural networks, including those used for time series forecasting, follows the same operational pipeline: repeated prediction, evaluation, and correction. This procedure progressively optimizes the network parameters until satisfactory accuracy is achieved or problems associated with excessive adaptation to the data emerge.

3.4.4 Additional Architectures

Although feedforward networks (FFNNs/MLPs) are the simplest and most widespread model, more sophisticated architectures have been developed in recent years. They are particularly suitable for time series forecasting because they capture medium- and long-term temporal dependencies and complex patterns. The main architectures are:

- *Recurrent Neural Networks (RNNs)*
RNNs extend traditional neural networks by introducing recurrent connections between neurons, enabling the model to

remember information from earlier time points. This makes them particularly suitable for sequential data such as time series. However, RNNs suffer from numerical instability, including vanishing and exploding gradients, which limits their effectiveness for very long sequences.

- LSTM and GRU (Long Short-Term Memory and Gated Recurrent Unit)

LSTM and GRU architectures extend RNNs to overcome their structural limitations. Both introduce internal control mechanisms called gates that regulate information flow and selectively remember or forget relevant information over time. LSTMs use three gates, input, output, and forget, along with a dedicated memory cell, allowing them to model long-term relationships such as trends and economic cycles effectively. GRUs are a more compact and computationally efficient variant, combining some of these gates while maintaining comparable performance. Both architectures are now among the most widely used for forecasting economic and financial time series, where capturing nonlinear patterns and long-lasting dependencies is essential.

- Convolutional Neural Networks (CNNs)

Although originally developed for image processing, CNNs have also been applied to time series. Their convolutional filters automatically extract local patterns and meaningful features from data, making them particularly useful in combination with RNNs or LSTMs.

3.5 Alternative Methods

3.5.1 Introduction

Increasingly hybrid and advanced approaches constitute the current state of research in time series analysis and forecasting. Alongside classical methods such as ARIMA and ETS, architectures have been developed that combine statistical, probabilistic, and deep learning components, offering a more realistic and flexible representation of temporal dynamics.

These methods integrate heterogeneous models: Hidden Markov Models (HMMs), which describe a series as a sequence of latent states; deep hybrid architectures such as DLS-TS-Net, which combine CNNs, LSTMs, GRUs, and autoregressive components; and more recent foundation models, based on self-attention mechanisms and large-scale pretraining.

Their common objective is to improve forecasting capability and robustness by combining the theoretical foundations of statistical models with the adaptive flexibility of deep learning. These approaches represent the current frontier of forecasting research, moving toward models that can independently learn the complexity and variability of temporal phenomena.

3.5.2 Hidden Markov Model Applications

Hidden Markov Models (HMMs) are among the most established probabilistic methods for modeling time series characterized by latent regimes or states: variables that cannot be observed directly but influence the behavior of the data over time. The underlying idea is that the observed phenomenon is generated by a hidden process that evolves according to a Markov chain, where each state depends only on the preceding state and each observation is generated according to the current state.

In practice, an HMM describes a time series as the outcome of two related processes:

1. A hidden process representing the system's internal states, such as a bull, or expansionary, phase and a bear, or recessionary, phase in an economic context.
2. An observed process corresponding to the empirical data, which depends probabilistically on the hidden state.

This dual structure models complex dynamics that purely linear models could not capture, while providing an interpretable representation of transitions between different behavioral regimes.

HMM parameters are identified iteratively using the Baum-Welch algorithm, based on Expectation-Maximization (EM). This algorithm uses the forward-backward procedure to estimate transition probabilities between hidden states and emission probabilities associated with observations. The Viterbi algorithm determines the most likely state sequence. Selection of the optimal number of states and assessment of model fit rely on information criteria such as AIC and BIC, discussed in earlier chapters.

Their flexible structure makes HMMs useful in quantitative finance, pattern recognition, bioinformatics, and dynamic system monitoring, particularly when data are noisy or nonstationary.

Recent research has introduced extensions that improve HMM flexibility and forecasting capability. Ordinal HMMs, for example, handle ordinal series whose observations represent ordered levels or categories, such as low, medium, and high, improving model parsimony and interpretability. Hybrid models integrating HMMs with deep learning techniques, such as neural networks and autoencoders, have also been developed. In these approaches, the neural network learns continuous representations of the data, while the HMM models latent sequential dependencies. Integrating the two through ensemble learning, using several simple learners to produce a final forecast that is more accurate and robust than a single model's forecast, improves robustness and generalization while reducing computational complexity.

HMMs therefore provide a conceptual bridge between classical statistical models and machine learning methods, enabling temporal processes with

hidden states or structural discontinuities to be modeled. Recent extensions, such as ordinal HMMs and neural network-HMM hybrids, confirm their central role in modern time series analysis and their importance for modeling complex nonlinear phenomena.

3.5.3 DLS-TS-Net

DLS-TS-Net (Deep Learning Structure for Time Series Networks) is one of the more recent and advanced architectures for multivariate time series forecasting. It was developed to overcome the limitations of classical models and early hybrid approaches, such as ARIMA-LSTM. The architecture combines convolutional, recurrent, and autoregressive models, using their complementary strengths to capture different levels of information in the data.

The general structure is divided into three parts:

- **CNN**

The convolutional component identifies local patterns and short-term dependencies in multivariate data. Convolutional filters move over the data sequence as sliding windows, establishing relationships within short, localized periods, such as shocks or microtrends.

- **LSTM and GRU**

The recurrent component consists of LSTM and GRU layers, two architectures derived from classical RNNs but designed to address long-term memory. LSTM retains relevant information for many time steps, making it particularly effective for capturing long-term trends and cycles. GRU is a more compact and computationally efficient version of LSTM, combining the functions of LSTM's three gates into two. This makes GRU suitable for medium-length time series or settings with limited data.

Their integration balances computational efficiency and memory capacity, producing a model that captures both long-term relationships and more immediate dynamics.

- Autoregressive Component (AR/VAR)

This final component captures direct relationships between past observations and future values, integrating them with the nonlinear information learned by the neural components.

The overall forecast combines the contributions of the different components:

$$\hat{y}_t = \widehat{y_{t,CNN}} + \widehat{y_{t,LSTM-GRU}} + \widehat{y_{t,AR}}$$

Each term represents the predictive contribution of the corresponding model component. The integration of CNN, LSTM, GRU, and AR gives DLS-TS-Net several advantages over traditional models and earlier hybrids:

- Structural Completeness: it combines linear and nonlinear components, capturing both local and global dependencies.
- Multivariate Capability: it allows multiple correlated series, such as indices, returns, and volatility, to be analyzed simultaneously.
- Robustness and Stability: the AR component mitigates numerical divergence and improves model stability.

DLS-TS-Net is therefore one of the more advanced solutions for modeling complex time series. Its deep hybrid architecture combines long-term memory, local pattern extraction, and linear consistency. In finance, it can provide reliable forecasts even in the presence of high volatility and dynamic correlations between variables, making it a reference point for modern deep forecasting applications.

3.5.4 Foundation Models

Foundation models are another particularly relevant research direction, extending the transformer concept. These models are pretrained on a very large collection of heterogeneous time series and then adapted, through fine-tuning, to a specific forecasting task using a much smaller dataset.

This pretraining and fine-tuning approach gives foundation models broad forecasting capability and enables transfer learning.

Pretraining on a large, diverse collection of time series allows a model to develop a deep understanding of general temporal patterns and dynamics, forming a knowledge base.

Fine-tuning also enables efficient adaptation: the pretrained model needs only a small amount of task-specific data to optimize its performance on the final task. This makes these models particularly valuable and efficient when time series data are scarce or in cold-start problems where traditional models would not generalize.

These models build on the concept of attention introduced in transformers, which allows the model to:

- Weight Importance: the model learns to dynamically weight the importance of different inputs or past internal states when processing the current element.
- Capture Long-Term Dependencies: when forecasting a value at time t , the model does not simply consider the immediately preceding time point, $t - 1$. It evaluates which past time points, such as $t - 10$ or the previous year's seasonal peak, are most relevant to the current forecast.

Emerging models and notable architectures in this area, often based on transformers, include TimeGPT, one of the first foundation models applied to time series; Lag-Llama; and Tiny Time Mixers, which can also handle

multivariate time series while using substantially fewer parameters than competing models.

Foundation models are a rapidly evolving research area. Time series modeling is no longer tied to a single approach, but draws on a collection of techniques that, when applied strategically, often in hybrid or ensemble combinations, offer powerful and versatile forecasting tools for addressing real-world complexity and capturing its underlying nonlinear relationships.

Chapter 4

Case Studies and Implementation

4.1 Introduction

This chapter describes and discusses forecasting systems implemented using several methods presented in this thesis. It considers two time series with different levels of complexity and concludes with a discussion of the results. In all experiments, visualizations were produced in Python using matplotlib, while NumPy and pandas were used for data management.

4.2 Data

All experiments were performed on a custom-built desktop computer. Computations were carried out on an AMD Ryzen 7 5700X CPU with eight physical cores and 32 GB of RAM.

4.2.1 Data for Directional Forecasting: The S&P 500 Index

The primary experiment focuses on directional forecasting accuracy in a financial setting. Historical data for the S&P 500 stock index were used as the dependent variable.

For modeling, logarithmic returns derived from closing prices were used for the HMMs for stability, while percentage returns were used for the other models and normalized for comparison. Data frequency differed to optimize each model's performance:

- HMM (Hidden Markov Model): trained on weekly data, resampled using W-FRI because trading runs from Monday to Friday, to reduce noise and facilitate the identification of stable market regimes, such as high and low volatility.
- Linear/Logistic Regression and ARIMA, including its variants: applied to daily data to capture short-term temporal dependencies, where these models perform best, while also using W-FRI resampling.

The analysis uses a rigorous walk-forward methodology with a fixed two-year training window, approximately 504 trading days. This configuration prevents data leakage and provides a robust out-of-sample evaluation.

The forecast, or test, periods vary according to each model's computational requirements:

- HMM and Linear/Logistic Regression: because of their relatively fast fitting and weekly partitioning, walk-forward analysis can cover a longer period, implemented from approximately 2005 onward.
- ARIMA: this model requires optimization of the (p, d, q) orders at every walk-forward iteration. Because of this high computational cost, ARIMA's walk-forward analysis is restricted to more recent windows, starting in 2020. This concentrates resources on the most recent historical period while retaining a rigorous methodological comparison.

To enrich the modeling, the following exogenous variables were retrieved through yfinance and incorporated: T-bill yield (10-year), volatility (VIX), WT-Brent prices, and inflation. All data were managed in pandas using its standard DataFrame structure for time series.

4.2.2 Univariate Analysis: European Flights

This section evaluates the forecasting accuracy of classical univariate statistical models, ARIMA, ETS, and SARIMA, on a time series with strong seasonality and an exogenous shock, COVID-19. This experiment provides a very-short-term forecasting benchmark for comparison with directional forecasting methods, demonstrating greater forecasting capability on a dataset that is more conventional and simpler than the preceding one.

The analysis uses European air traffic data, aggregated daily flight counts, from 2016 onward. The dataset is particularly relevant because of its pronounced seasonality and the profound structural shock caused by the COVID-19 pandemic. The objective is to quantify how well the models forecast the period immediately following training when the data exhibit seasonality and other conventional characteristics.

The backtesting approach uses a static chronological train/test split:

- Training Set: historical data up to the days preceding the end of the series.
- Test Set (Out of Sample): strictly the final 12 days of the time series.

This narrow test window restricts the analysis to very-short-term forecasting and does not represent medium- or long-term forecasting performance.

Forecasting performance is measured using continuous error metrics, including RMSE/MSE, MAE, MAPE, and R-squared, all introduced earlier. These are standard metrics for evaluating predictions against the 12-day test set. Here too, data are managed in pandas using the DataFrame structure.

4.3 Technical Implementation: S&P 500

This section describes the practical implementation of the models used to forecast the financial dataset and the Walk-Forward Analysis (WFA) methodology at the heart of the out-of-sample stock return analysis. All models were implemented in Python.

4.3.1 Hidden Markov Model (HMM) Architecture

For HMMs, the main objective is to identify distinct periods, or states, within the series. Several configurations were tested with two or three hidden states, representing high- and low-volatility conditions. A slightly more advanced strategy was also implemented for comparison, using multiple Gaussian distributions to capture returns. According to the literature, stock returns do not follow a standard Gaussian distribution, but exhibit fat tails and other distinctive features. Using three Gaussian distributions provides better coverage of the full range of returns. In practice, the Gaussian HMM implementation is available in the `hmmlearn` library.

Forecasting then uses a WFA approach with a training window of 104 or 156 weeks. After training, the most likely state for the following week is calculated. The data for predicting t_{test} are partitioned as follows:

- Training Data: from $t_{test} - \text{WindowSize}$ to $t_{test} - 1$
- Test Data: the single point t_{test} , corresponding to the week to be forecast.

This strategy leaves substantial room for improvement through further hyperparameter optimization and the possible addition of correlated exogenous variables, such as volatility or sentiment analysis measures.

4.3.2 ARIMA/ARIMAX Architecture

The statsmodels library was selected to implement ARIMA models. ARIMA follows the standard formulation described in Chapter 2. The only modifications concern techniques used to optimize computation:

- Dynamic selection of the order d using the adfuller stationarity test, followed by adjustment through differencing.
- A restricted search space for p and q to accelerate convergence to the minimum AIC.
- Fewer iterations than in the standard configuration.

These optimizations are essential when generating daily forecasts using the same WFA approach. For the same reasons, the test period is also restricted to 2020 onward.

ARIMAX uses a comparable implementation, with the addition of exogenous variables such as the VIX index, US10Y, the 10-year U.S. Treasury yield, and the price of Brent crude oil.

4.3.3 Regression Architecture

Regression was implemented using sklearn, with multiple auxiliary variables as in ARIMAX, but using indicators derived from the S&P 500 itself, such as moving averages, variance, and SMA.

The approach is comparable to the preceding ones and uses the same WFA methodology, with a two-year window and daily point forecasts. For linear regression, which predicts continuous values, the sign of the forecast determines the predicted direction, while its absolute value is used as an estimate of confidence.

For logistic regression, the direct prediction is used because it is binary.

4.4 Technical Implementation: European Flights

This section describes the practical implementation of models applied to the European flights dataset from 2016 onward. It serves as a case study for forecasting real-world time series with less complex seasonality and variability. The experiment evaluates model forecasting capability and robustness in a nonfinancial setting, without the nonlinearities and complexity of the preceding financial case. All procedures were developed in Python using statistical libraries.

The implementations are directly comparable across the selected models. All three use the previously mentioned statsmodels library, which includes the specific functions required for this set of models.

The following implementations were selected:

- ARIMA, using the dedicated ARIMA function.
- ETS, using the ExponentialSmoothing function.
- SARIMA, using SARIMAX without specifying exogenous variables.

The models use a conventional train/test split, with the final 12 days of the time series as the test set. Orders and parameters were freely selected and evaluated through repeated for-loop searches to minimize AIC for ARIMA and identify seasonality and trend for ETS. No computational constraints or optimizations were imposed because they were unnecessary for such a short forecast horizon.

4.5 Results

4.5.1 S&P 500

The following charts present the results, followed by a table summarizing the metrics.



Figure 4.1.1: Simple HMM with two states and a single Gaussian distribution



Figure 4.1.2: Simple HMM with three states and a single Gaussian distribution

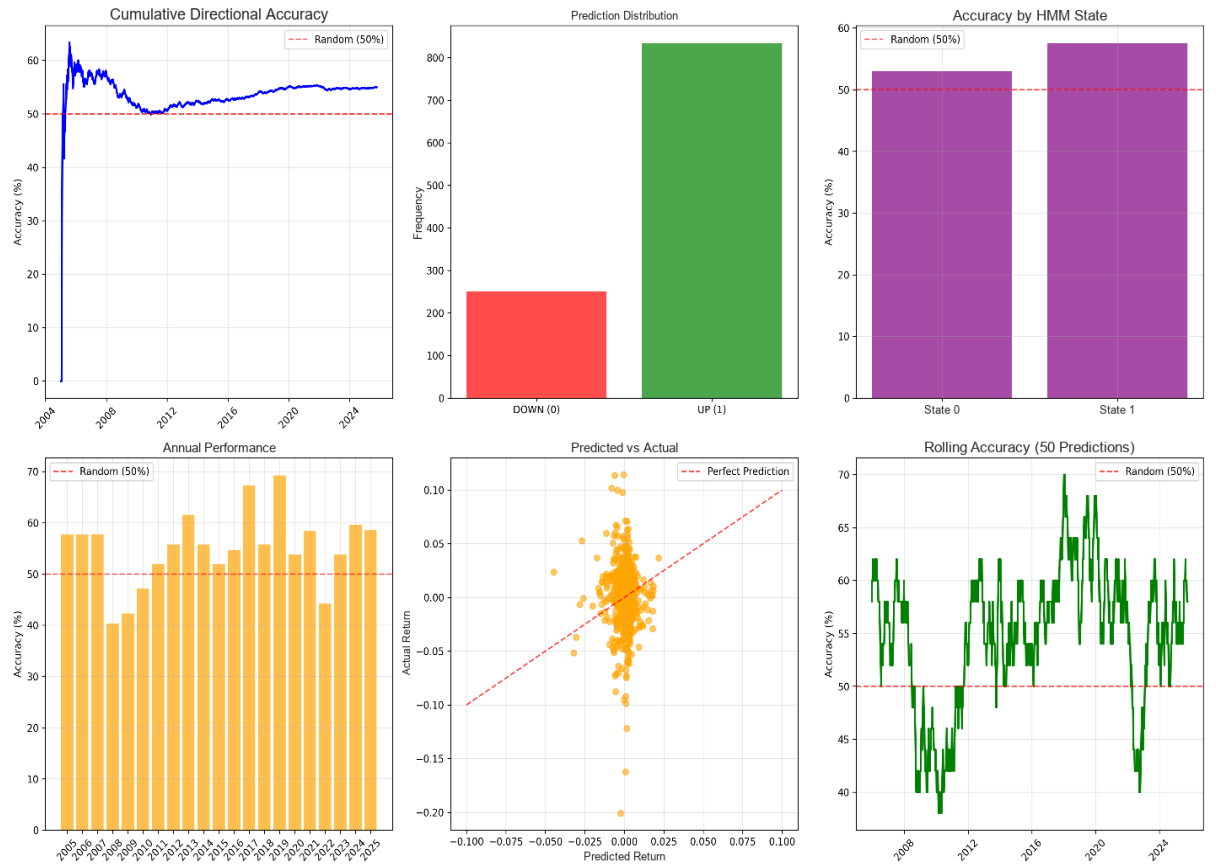


Figure 4.1.3: HMM with two states and three Gaussian distributions to model returns



Figure 4.1.4: Standard ARIMA with a rolling window (reduced time frame)

Figure 4.1.5: ARIMAX with four exogenous variables (VIX, Brent, US10Y, Gold), selecting the best three at each iteration

ARIMAX Walk-Forward Analysis - Directional Prediction Results



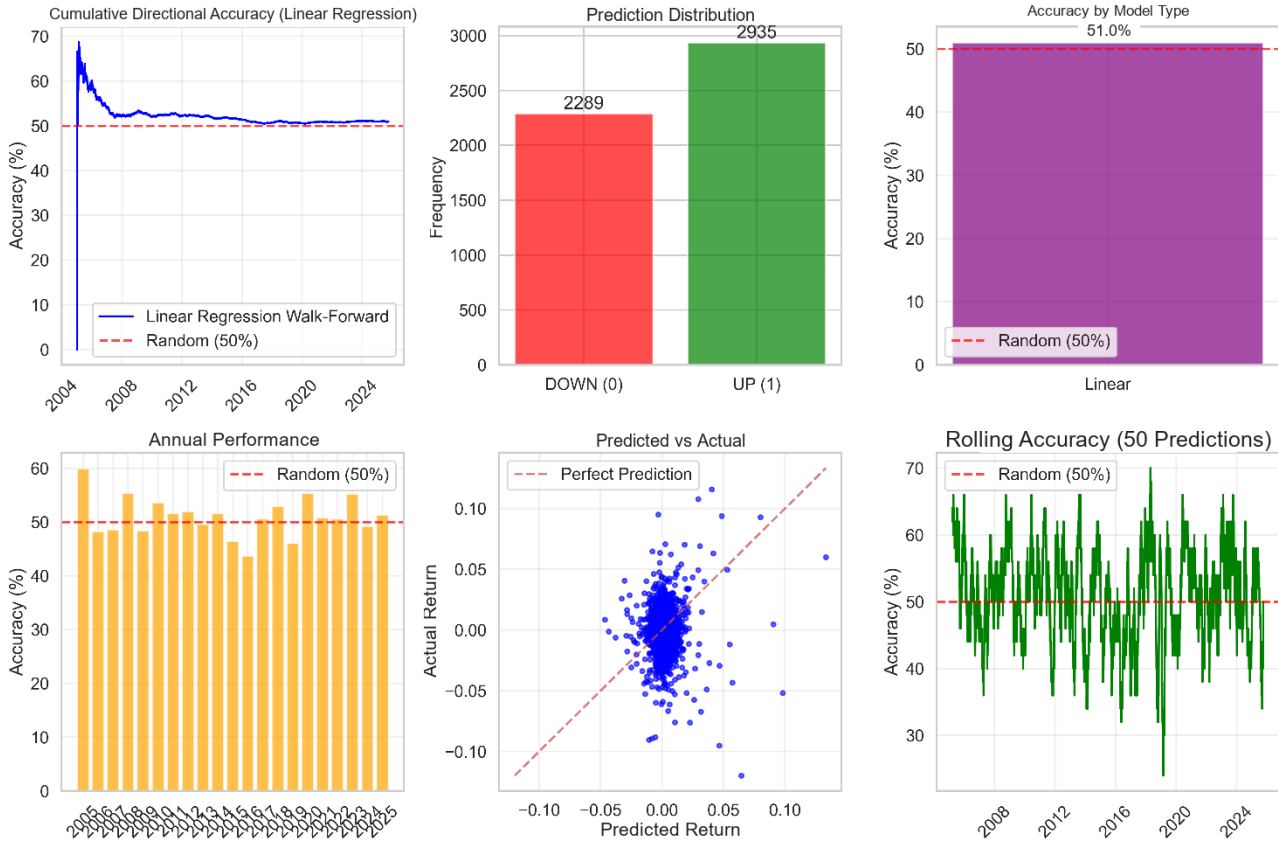


Figure 4.1.6: Linear regression with exogenous variables derived from the S&P 500 dataset

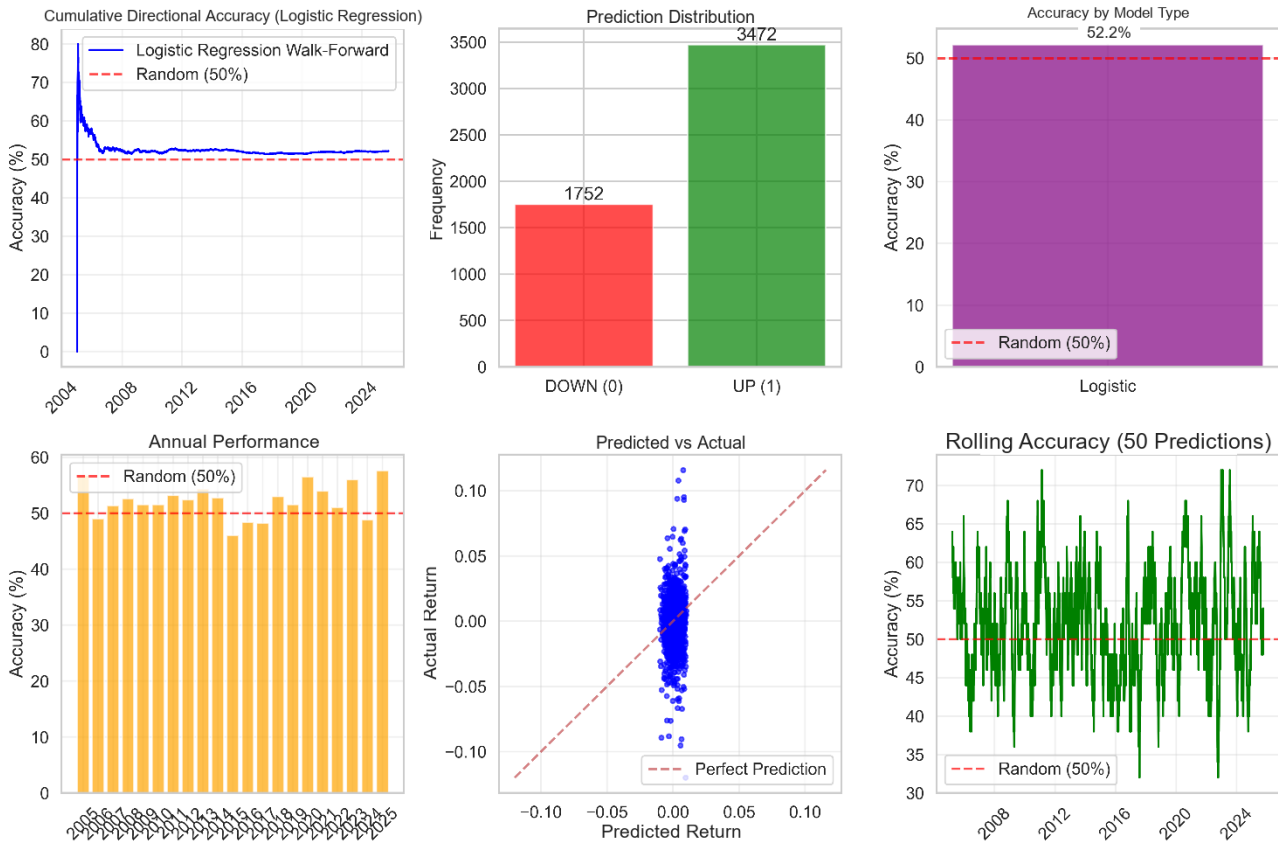


Figure 4.1.7: Logistic regression using the same exogenous variables

To conclude the experiment, the table reports the overall Directional Accuracy of the models. The HMM approach performs best, while still leaving substantial room for improvement.

These results should be interpreted against the 50% accuracy baseline associated with a coin toss. Even a few percentage points above this random baseline can provide a meaningful statistical edge.

Some inaccuracies and unstable predictions can be attributed to factors that the models do not capture, such as geopolitical crises, sentiment noise, potential financial bubbles, and dividend payments. These should be considered in more detailed analyses. The results should not be interpreted as financial advice or as evidence of effective trading strategies, because many real-world factors remain unmodeled, including transaction costs and fees that could entirely erode the statistical edge.

Model	Cumulative Accuracy	2025 Accuracy	UP Accuracy
HMM (2 states)	53.97%	53.66%	59.32%
HMM (3 states)	52.91%	48.78%	56.70%
HMM-Adv (2 states)	55.07%	58.54%	57.96%
ARIMA	52.08%	52.88%	54.77%
ARIMAX	52.23%	52.86%	54.48%
Logistic-Reg	52.16%	57.59%	55.07%
Linear-Reg	51.00%	51.31%	54.96%

The results indicate that the two-state HMM modeling returns with three Gaussian distributions achieves the largest cumulative edge. Even an advantage of approximately five percentage points over the baseline suggests practical potential. With substantial room for improvement, this is a promising result. It is also interesting to observe the near inability of linear regression to adapt to such complex and unstable data. ARIMAX offers a slight improvement

by modeling exogenous variables; a more careful selection of these variables could further improve its performance.

4.5.2 European Flights

This section presents the results of the experiment forecasting the number of flights in Europe over a 12-day period.

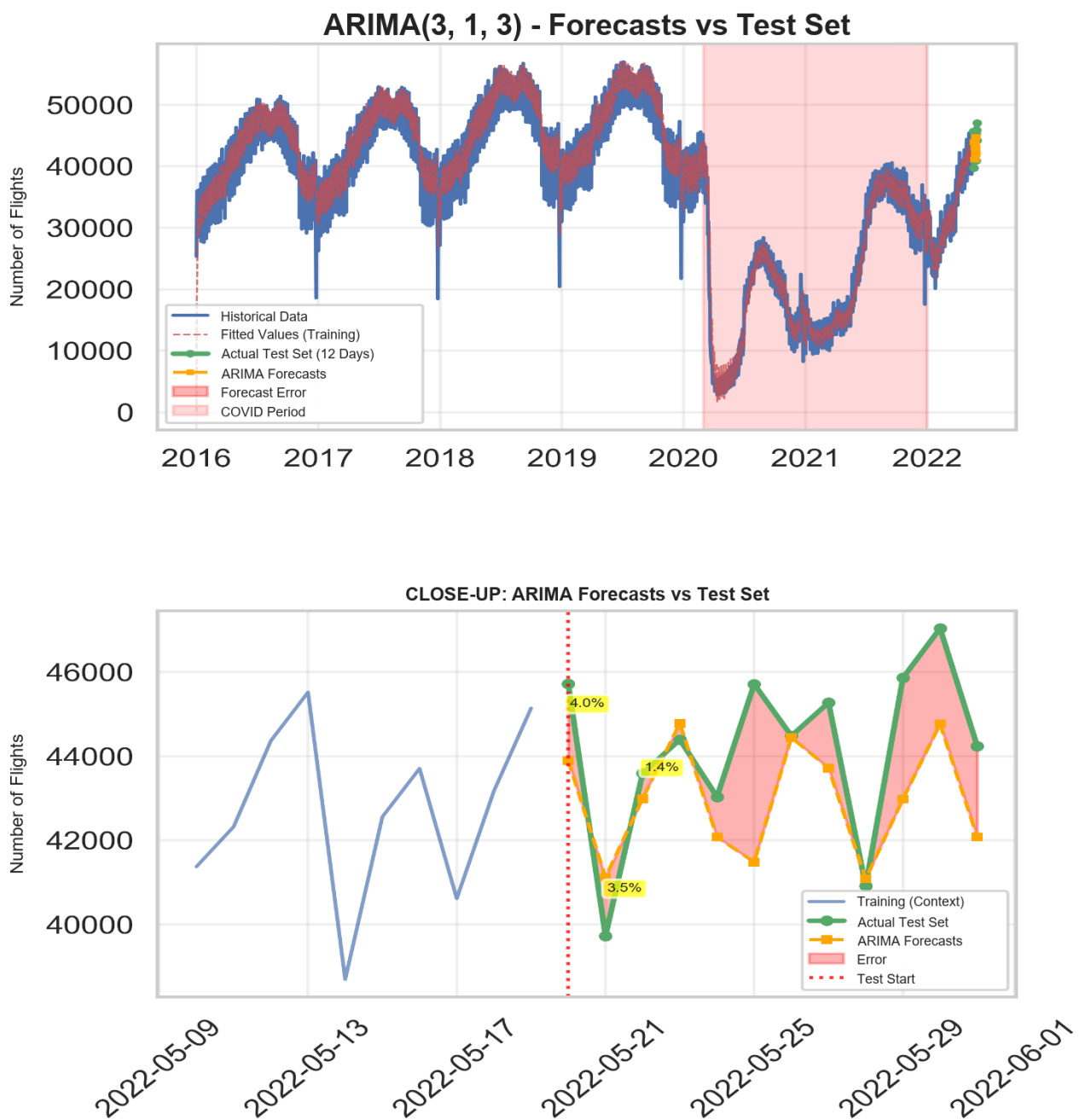
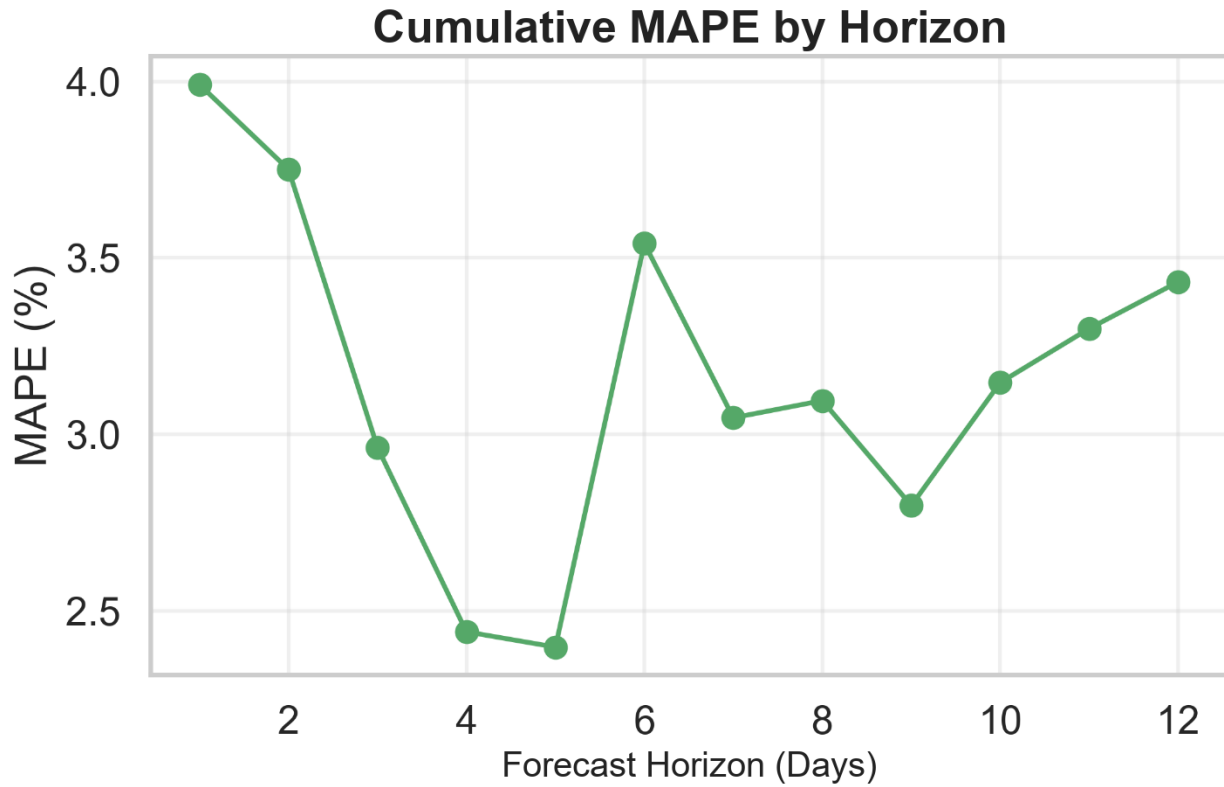
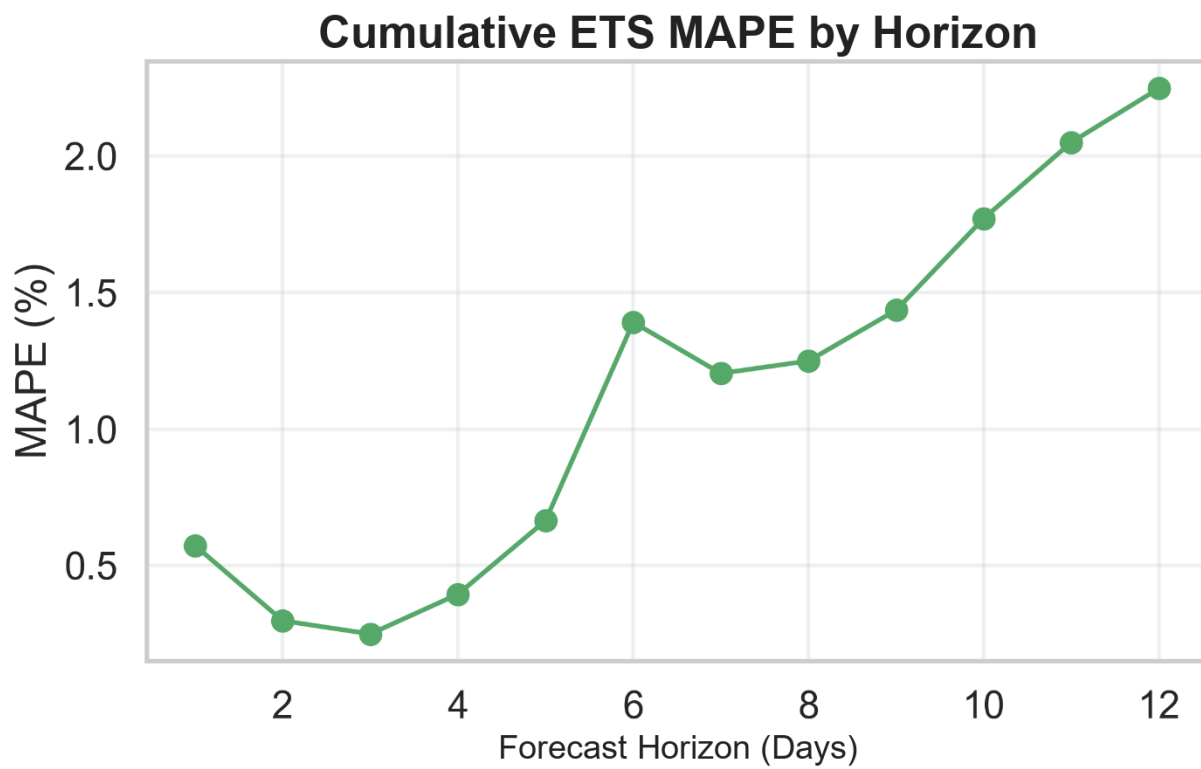


Figure 4.2.1: ARIMA forecast over the 12-day period

*Figure 4.2.2: Cumulative MAPE for ARIMA**Figure 4.2.3: Cumulative MAPE for ETS*

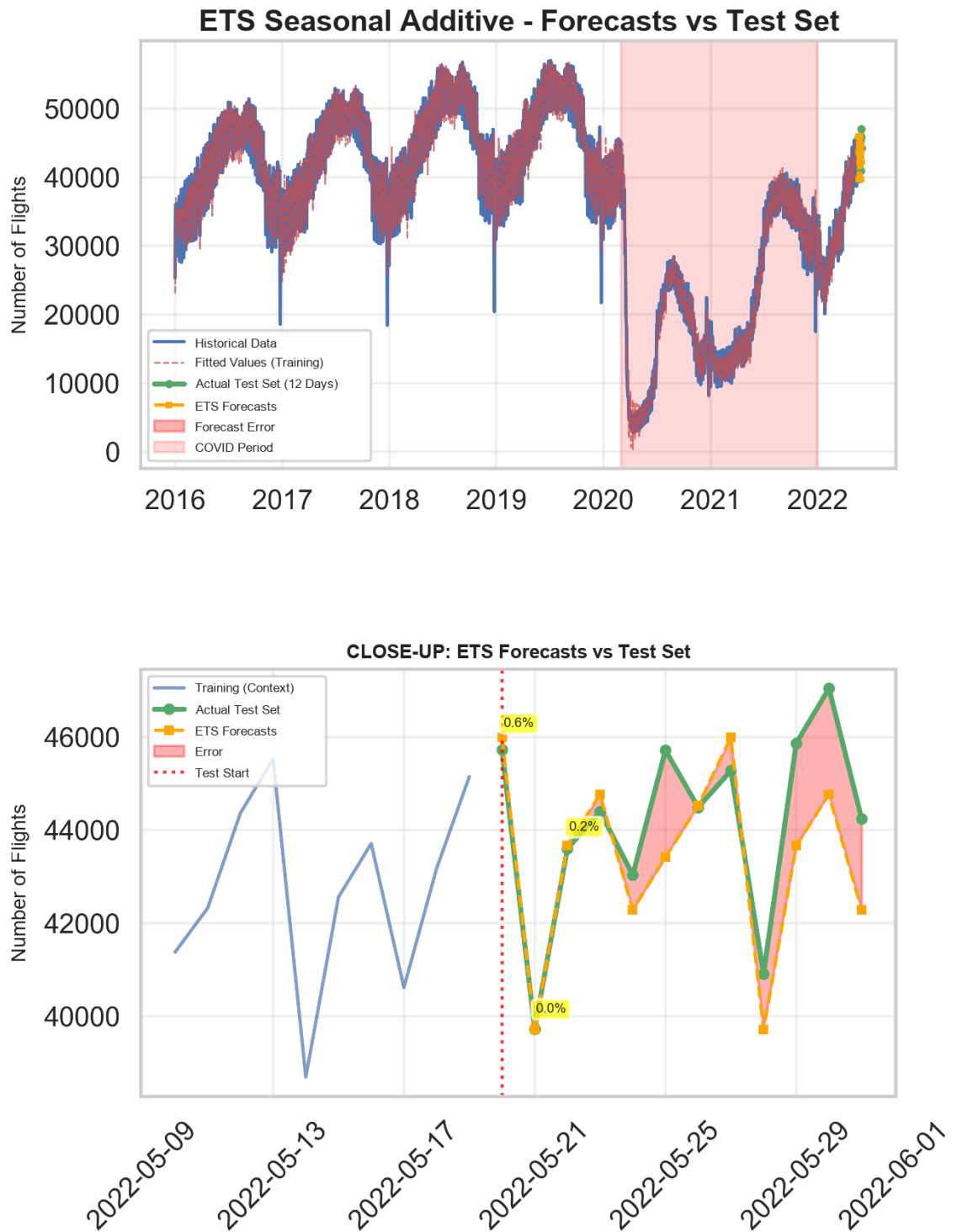


Figure 4.2.4: ETS forecast over the 12-day period

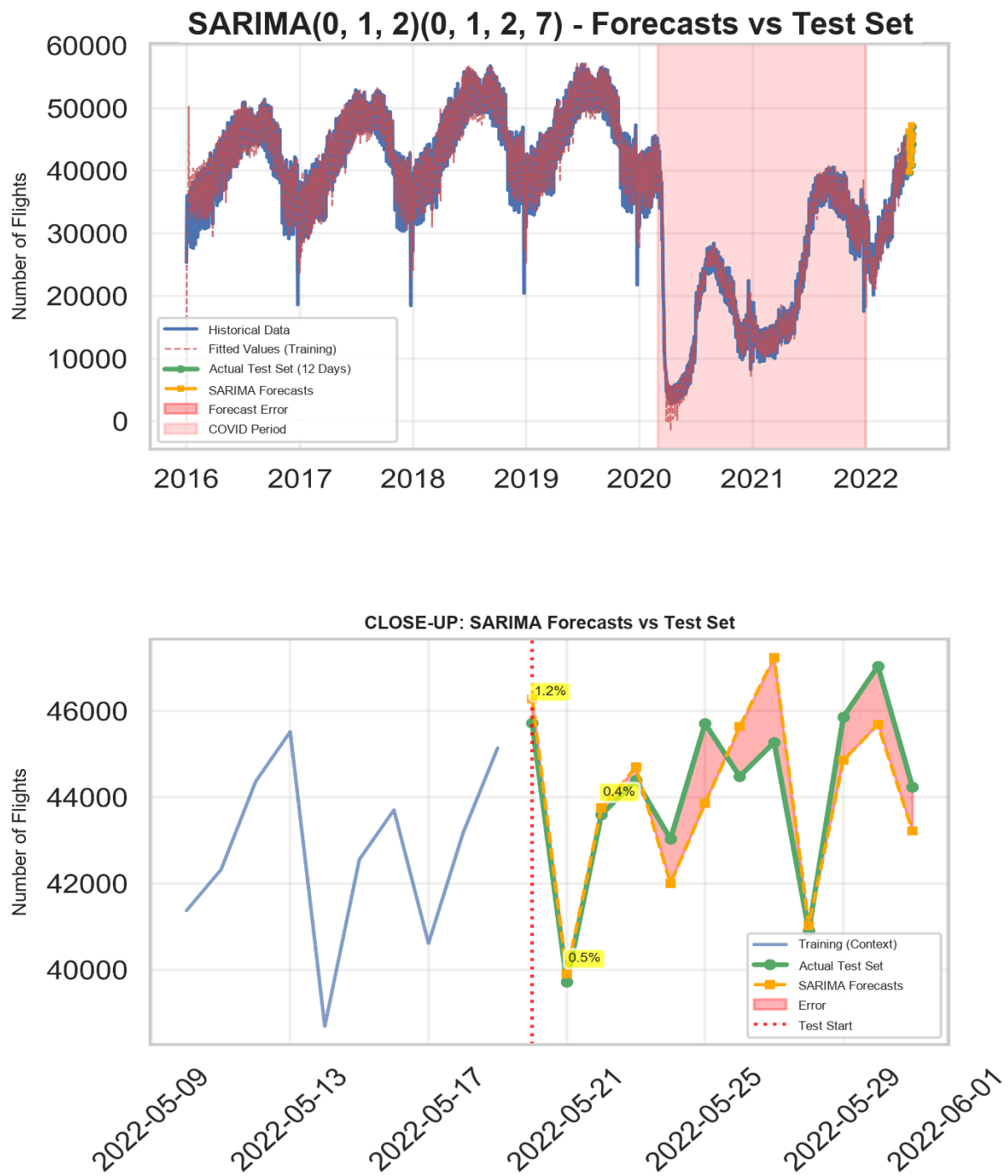


Figure 4.2.5: SARIMA forecast over the 12-day period

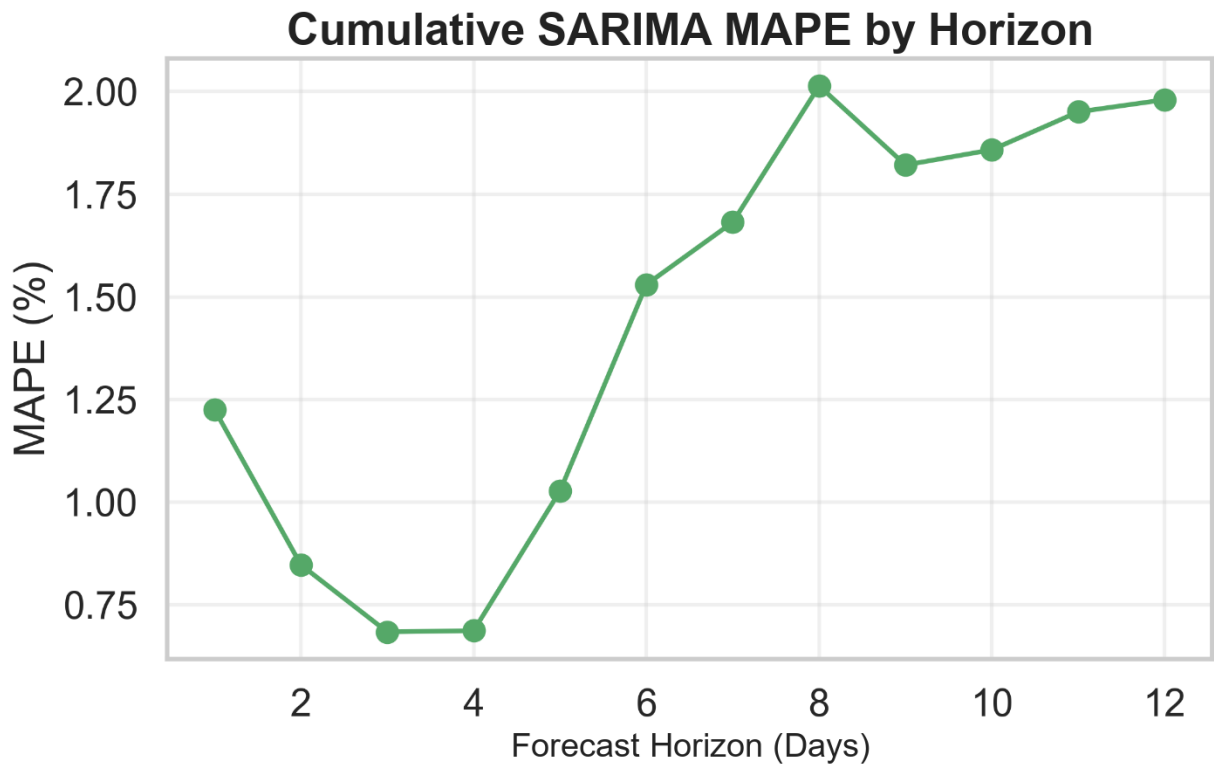


Figure 4.2.6: Cumulative MAPE for SARIMA

The results show that SARIMA performs best because it adapts more effectively to the seasonality of the time series. The additional metrics in the following table provide an overall view of model performance. SARIMA ranks first, followed by ETS. Better coefficient selection could potentially improve ETS performance further.

Greater computing power and libraries optimized for parallel GPU computation could substantially improve computational performance and, consequently, forecasting quality by allowing more auxiliary variables and a broader window for parameter selection.

Model	MAPE	R-Squared	RMSE (flight count)
ETS	2.25%	0.56	1349
ARIMA	3.43%	0.083	1957
SARIMA	1.98%	0.71	1081

Chapter 5

Conclusions

This thesis aimed to provide a comprehensive overview of the main methods for time series analysis and forecasting.

Its purpose was to offer a solid, conceptually clear foundation for further study of individual models and their distinctive characteristics, rather than develop a highly formal mathematical treatment.

The experiments also highlighted the strengths and limitations of the different approaches and their adaptability to the nature of the data. In particular, implementation on two heterogeneous datasets demonstrated that each model responds differently to specific characteristics of the series, confirming the importance of informed and targeted methodological choices.

This result is especially relevant given the complexity and inherent variability of real-world phenomena, particularly in finance, where data are often dominated by inconsistencies, inefficiencies, and sudden shocks whose origins cannot readily be identified quantitatively.

From this perspective, time series analysis can be compared to the study of human behavior: a process that seeks both understanding and the ability to anticipate future developments.

Finally, the experimental evidence indicates substantial opportunities for improvement. Greater computing power, optimized pipelines, and parallel or distributed computing techniques could enable more efficient processing of larger datasets and significantly improve forecasting quality. The work presented here therefore provides a starting point for developing models that are increasingly robust, adaptive, and capable of learning real-world complexity.

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